

Technical Documentation for the Rb87 Bias-Coil and 3D MOT Gradient-Field Simulation

MIGA Program

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Contents

1	Purpose of This Document	3
2	Code Structure and Main Files	3
3	Coordinate System and Geometry	3
3.1	Atomic-Cloud Position	3
3.2	Three-Axis Compensation Coil Pairs	4
3.3	3D MOT Gradient-Coil Pair	4
4	Compensation-Coil Field Model	5
4.1	Biot–Savart Law	5
4.2	Current-to-Field Matrix	5
5	3D MOT Gradient-Field Model	5
5.1	Anti-Helmholtz Circular Pair	5
5.2	Local Gradient Tensor	6
6	Total Magnetic Field During Optical Molasses	6
7	Cooling-Efficiency Model	7
7.1	Magnetic Sensitivity	7
7.2	Optical Pumping Rate	7
7.3	Effective Magnetic Width	7
7.4	Relative Cooling Efficiency	7
8	Temperature-Evolution Model	8
8.1	Field-Dependent Equilibrium Temperature	8

8.2	Field-Dependent Cooling Time Constant	8
8.3	Temperature Dynamics	8
9	Current Scan and Local Refinement	8
9.1	Coarse Scan	8
9.2	Local Refinement	9
10	Local Spatial Field Curves	9
11	Representative Figures and Their Interpretation	9
11.1	Overview Scan Figure	10
11.2	Time-Domain Dynamics Figure	11
11.3	Local Spatial Field Figure	12
12	Configuration and Output Files	12
13	Graphical User Interface	13
14	Model Scope and Limitations	13
15	Possible Future Extensions	14
16	Conclusion	14

1 Purpose of This Document

This document describes the functionality, physical assumptions, and governing formulas of the simulation code centered on `rb87_bias_coils_current_scan.py`. The program is intended for Rb87 cold-atom experiments and focuses on the optical molasses / polarization-gradient-cooling (PGC) stage after the 3D MOT magnetic field is switched off.

The present version of the code is designed to answer the following practical questions:

- How does the final molasses temperature depend on the compensation currents in the three bias-coil pairs?
- How does the answer change if the atomic cloud is not exactly located at the magnetic-field zero?
- How does the residual 3D MOT quadrupole field influence the compensation effectiveness when the cloud is displaced from the origin?
- How do the compensation coils modify not only the field at the cloud center, but also the local field profile around the cloud?

2 Code Structure and Main Files

File	Role
<code>rb87_bias_coils_current_scan.py</code>	Main simulation core. Contains coil-field calculation, MOT gradient-field calculation, temperature evolution, current scan, local refinement, plot generation, and summary export.
<code>rb87_bias_coils_current_scan_gui.py</code>	Qt desktop interface for interactive parameter tuning and figure inspection.
<code>rb87_bias_coils_current_scan_tk_gui.py</code>	Tk fallback interface used when Qt is unavailable or unstable.
<code>example_config.json</code>	Example configuration file containing atomic, molasses, bias-coil, MOT-gradient-coil, scan, and spatial-probe parameters.
<code>outputs/</code>	Output folder containing CSV tables, PNG figures, summary files, and resolved configuration files.

3 Coordinate System and Geometry

3.1 Atomic-Cloud Position

The optical molasses stage is evaluated at the center of the atomic cloud,

$$\mathbf{r}_0 = (x_0, y_0, z_0), \tag{1}$$

with coordinates expressed in mm. In the configuration file this is written as

$$\text{molasses_position_mm} = [x_0, y_0, z_0]. \quad (2)$$

By default,

$$\mathbf{r}_0 = (0, 0, 0), \quad (3)$$

which means that the cloud center coincides with the geometric center of the compensation-coil arrangement and the nominal zero of the MOT quadrupole field.

3.2 Three-Axis Compensation Coil Pairs

The compensation system consists of three square coil pairs aligned with the laboratory axes X , Y , and Z . Each axis is modeled as one symmetric pair of square loops:

- each coil has N_{comp} turns,
- each loop has side length L ,
- the two loop centers are placed at $\pm d_{\text{comp}}$ along the corresponding axis,
- the two coils in one pair produce the same field direction near the center.

The current default geometry is

$$N_{\text{comp}} = 15, \quad L = 30 \text{ cm}, \quad d_{\text{comp}} = 20 \text{ cm}. \quad (4)$$

3.3 3D MOT Gradient-Coil Pair

The simulation also contains a dedicated model for the 3D MOT magnetic coils. They are represented as a circular anti-Helmholtz pair:

- each coil has N_{MOT} turns,
- each coil has radius R_{MOT} ,
- the two coil centers are located at $\pm d_{\text{MOT}}$ along a chosen axis,
- the two coil currents have opposite sign, producing a near-zero field and non-zero gradient at the origin.

In the current example configuration,

$$N_{\text{MOT}} = 50, \quad R_{\text{MOT}} = 10 \text{ cm}, \quad d_{\text{MOT}} = 10 \text{ cm}, \quad I_{\text{MOT}} = 7 \text{ A}. \quad (5)$$

The coil-pair axis is taken to be parallel to the line $y = z$, that is,

$$\hat{\mathbf{u}}_{\text{MOT}} \parallel (0, 1, 1). \quad (6)$$

Internally the program normalizes this direction:

$$\hat{\mathbf{u}}_{\text{MOT}} = \frac{(u_x, u_y, u_z)}{\sqrt{u_x^2 + u_y^2 + u_z^2}}, \quad (7)$$

which for the default case becomes

$$\hat{\mathbf{u}}_{\text{MOT}} = \frac{(0, 1, 1)}{\sqrt{2}}. \quad (8)$$

4 Compensation-Coil Field Model

4.1 Biot–Savart Law

The compensation-coil field is calculated from the Biot–Savart law,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint \frac{d\boldsymbol{\ell}' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}. \quad (9)$$

Rather than using a purely on-axis center formula everywhere, the program discretizes each square loop into finite straight segments and evaluates the field numerically. This is important because once the cloud is displaced from the origin, off-diagonal coupling terms naturally appear.

4.2 Current-to-Field Matrix

At the cloud center $\mathbf{r}_0$, the compensation-coil contribution is written as

$$\mathbf{B}_{\text{comp}}(\mathbf{r}_0) = \mathbf{M}_{\text{comp}}(\mathbf{r}_0) \mathbf{I}_{\text{comp}}, \quad (10)$$

where

$$\mathbf{I}_{\text{comp}} = \begin{bmatrix} I_x \\ I_y \\ I_z \end{bmatrix}, \quad (11)$$

and

$$\mathbf{M}_{\text{comp}}(\mathbf{r}_0) = \begin{bmatrix} M_{xx} & M_{xy} & M_{xz} \\ M_{yx} & M_{yy} & M_{yz} \\ M_{zx} & M_{zy} & M_{zz} \end{bmatrix}. \quad (12)$$

When $\mathbf{r}_0 = (0, 0, 0)$ and the geometry is perfectly symmetric, the matrix is approximately diagonal:

$$\mathbf{M}_{\text{comp}}(0) \approx \begin{bmatrix} M_0 & 0 & 0 \\ 0 & M_0 & 0 \\ 0 & 0 & M_0 \end{bmatrix}. \quad (13)$$

For the current default geometry, the code yields approximately

$$M_0 \approx 296.35 \text{ mG/A}. \quad (14)$$

5 3D MOT Gradient-Field Model

5.1 Anti-Helmholtz Circular Pair

The 3D MOT field is modeled as the superposition of two circular coils with opposite currents. If the coil centers are

$$\mathbf{r}_{\pm} = \pm d_{\text{MOT}} \hat{\mathbf{u}}_{\text{MOT}}, \quad (15)$$

then the field at position $\mathbf{r}$ is written as

$$\mathbf{B}_{\text{MOT}}(\mathbf{r}) = \mathbf{B}_o(\mathbf{r}; \mathbf{r}_+, +I_{\text{MOT}}) + \mathbf{B}_o(\mathbf{r}; \mathbf{r}_-, -I_{\text{MOT}}), \quad (16)$$

where $\mathbf{B}_o$ denotes the field from one circular loop.

Because the two currents have opposite sign, the origin is close to a magnetic-field zero:

$$\mathbf{B}_{\text{MOT}}(\mathbf{0}) \approx \mathbf{0}, \quad (17)$$

while the gradient is non-zero.

5.2 Local Gradient Tensor

The code also evaluates the local Jacobian of the MOT field near the cloud center,

$$\mathbf{G}_{\text{MOT}}(\mathbf{r}_0) = \nabla \mathbf{B}_{\text{MOT}}(\mathbf{r}_0) = \begin{bmatrix} \frac{\partial B_x}{\partial x} & \frac{\partial B_x}{\partial y} & \frac{\partial B_x}{\partial z} \\ \frac{\partial B_y}{\partial x} & \frac{\partial B_y}{\partial y} & \frac{\partial B_y}{\partial z} \\ \frac{\partial B_z}{\partial x} & \frac{\partial B_z}{\partial y} & \frac{\partial B_z}{\partial z} \end{bmatrix}_{\mathbf{r}=\mathbf{r}_0}. \quad (18)$$

Numerically, the derivatives are approximated using central differences:

$$\frac{\partial \mathbf{B}}{\partial q} \approx \frac{\mathbf{B}(\mathbf{r}_0 + \Delta q \hat{\mathbf{e}}_q) - \mathbf{B}(\mathbf{r}_0 - \Delta q \hat{\mathbf{e}}_q)}{2\Delta q}. \quad (19)$$

6 Total Magnetic Field During Optical Molasses

The total field at the cloud center during molasses is modeled as

$$\mathbf{B}_{\text{tot}}(t, \mathbf{r}_0) = \mathbf{B}_{\text{stray}} + \mathbf{B}_{\text{comp}}(\mathbf{r}_0) + e^{-t/\tau_{\text{off}}} \mathbf{B}_{\text{switch}} + e^{-t/\tau_{\text{off}}} \mathbf{B}_{\text{MOT, res}}(\mathbf{r}_0). \quad (20)$$

Here:

- $\mathbf{B}_{\text{stray}}$ is the static background stray field,
- $\mathbf{B}_{\text{comp}}(\mathbf{r}_0)$ is the compensation field at the cloud center,
- $\mathbf{B}_{\text{switch}}$ is an empirical switch-off residual field,
- $\mathbf{B}_{\text{MOT, res}}(\mathbf{r}_0)$ is the local residual field from the MOT gradient pair,
- τ_{off} is the switch-off decay time constant.

The code introduces a configurable scale factor s_{MOT} for the MOT residual:

$$\mathbf{B}_{\text{MOT, res}}(\mathbf{r}_0) = s_{\text{MOT}} \mathbf{B}_{\text{MOT}}(\mathbf{r}_0). \quad (21)$$

Therefore, if the atomic cloud is exactly at the magnetic zero, the MOT residual term can remain very small. If the cloud is displaced, however, the local quadrupole field becomes non-zero and can significantly alter the early molasses dynamics.

7 Cooling-Efficiency Model

7.1 Magnetic Sensitivity

The effective Zeeman sensitivity is written as

$$\chi_B = |g_F| \frac{\mu_B}{h}, \quad (22)$$

converted in the code to units of Hz/mG.

7.2 Optical Pumping Rate

The total saturation parameter is taken as

$$s_{\text{tot}} = N_{\text{beam}} s_0, \quad (23)$$

where s_0 is the saturation parameter per beam and N_{beam} is the number of beams.

The optical pumping rate is approximated as

$$\Gamma_{\text{op}} = \frac{1}{2} \Gamma \frac{s_{\text{tot}}}{1 + s_{\text{tot}} + 4(\Delta/\Gamma)^2}, \quad (24)$$

where Γ is the natural linewidth and Δ is the laser detuning.

7.3 Effective Magnetic Width

Unless manually overridden, the code defines an effective magnetic width

$$B_{\text{width}} = \alpha_{\text{op}} \frac{\Gamma_{\text{op}}}{\chi_B}, \quad (25)$$

where α_{op} is the empirical width-scale factor corresponding to `optical_pumping_width_scale`.

7.4 Relative Cooling Efficiency

The suppression of cooling by residual magnetic field is modeled as

$$\eta_{\text{raw}}(t) = \frac{1}{1 + \left(\frac{|\mathbf{B}_{\text{tot}}(t, \mathbf{r}_0)|}{B_{\text{width}}} \right)^2}. \quad (26)$$

The code then applies a lower clipping threshold:

$$\eta(t) = \min(1, \max[\eta_{\text{min}}, \eta_{\text{raw}}(t)]). \quad (27)$$

8 Temperature-Evolution Model

8.1 Field-Dependent Equilibrium Temperature

Let T_0 be the zero-field limit temperature and T_{fail} be the high-field failure temperature. The instantaneous equilibrium temperature is written as

$$T_{\text{eq}}(t) = T_0 + (T_{\text{fail}} - T_0) [1 - \eta_{\text{raw}}(t)]. \quad (28)$$

Thus:

- for $|\mathbf{B}| \rightarrow 0$, one has $\eta_{\text{raw}} \rightarrow 1$ and therefore $T_{\text{eq}} \rightarrow T_0$;
- for $|\mathbf{B}| \gg B_{\text{width}}$, one has $\eta_{\text{raw}} \rightarrow 0$ and therefore $T_{\text{eq}} \rightarrow T_{\text{fail}}$.

8.2 Field-Dependent Cooling Time Constant

If τ_0 denotes the zero-field cooling time, the code defines

$$\tau_{\text{cool}}(t) = \frac{\tau_0}{\eta(t)}. \quad (29)$$

8.3 Temperature Dynamics

The temperature follows

$$\frac{dT}{dt} = -\frac{T(t) - T_{\text{eq}}(t)}{\tau_{\text{cool}}(t)}. \quad (30)$$

The implementation uses explicit Euler stepping:

$$T_{n+1} = T_n - \frac{T_n - T_{\text{eq},n}}{\tau_{\text{cool},n}} \Delta t. \quad (31)$$

The final temperature T_{final} is the temperature at the end of the molasses time window.

9 Current Scan and Local Refinement

9.1 Coarse Scan

The simulation first evaluates a user-defined three-dimensional current grid,

$$I_x \in [I_{x,\text{min}}, I_{x,\text{max}}], \quad I_y \in [I_{y,\text{min}}, I_{y,\text{max}}], \quad I_z \in [I_{z,\text{min}}, I_{z,\text{max}}]. \quad (32)$$

For every current triplet (I_x, I_y, I_z) , the code runs one full temperature-evolution simulation and stores:

- the final temperature,
- the cooling efficiency,
- the compensation field at the cloud center,
- the mean, initial, and final residual-field magnitudes.

9.2 Local Refinement

If refinement is enabled, the code automatically narrows the scan window around the best coarse point and attempts to reach a target current step

$$\Delta I_{\text{target}}. \quad (33)$$

This allows the workflow to preserve both:

- a sufficiently broad coarse scan, and
- a fine final overview around the optimum.

10 Local Spatial Field Curves

In addition to the time evolution at the cloud center, the code evaluates one-dimensional field cuts around the cloud. For a probe direction $\hat{\mathbf{e}}$, the sampled positions are

$$\mathbf{r}(s) = \mathbf{r}_0 + s \hat{\mathbf{e}}, \quad (34)$$

with

$$s \in [-s_{\text{max}}, s_{\text{max}}], \quad (35)$$

where s_{max} is controlled by the `spatial_probe` block.

The current implementation plots cuts along:

- the laboratory X direction,
- the laboratory Y direction,
- the laboratory Z direction,
- the MOT coil-pair axis $\hat{\mathbf{u}}_{\text{MOT}}$.

The resulting figures compare:

- the magnetic field from the 3D MOT gradient pair alone,
- the total field without compensation,
- the total field with the best compensation current.

This makes it possible to distinguish between pointwise compensation at the cloud center and a broader improvement of the field profile in the neighborhood of the cloud.

11 Representative Figures and Their Interpretation

The following figures are inserted directly from the current simulation outputs, so that the documentation is tied to the actual code products.

11.1 Overview Scan Figure

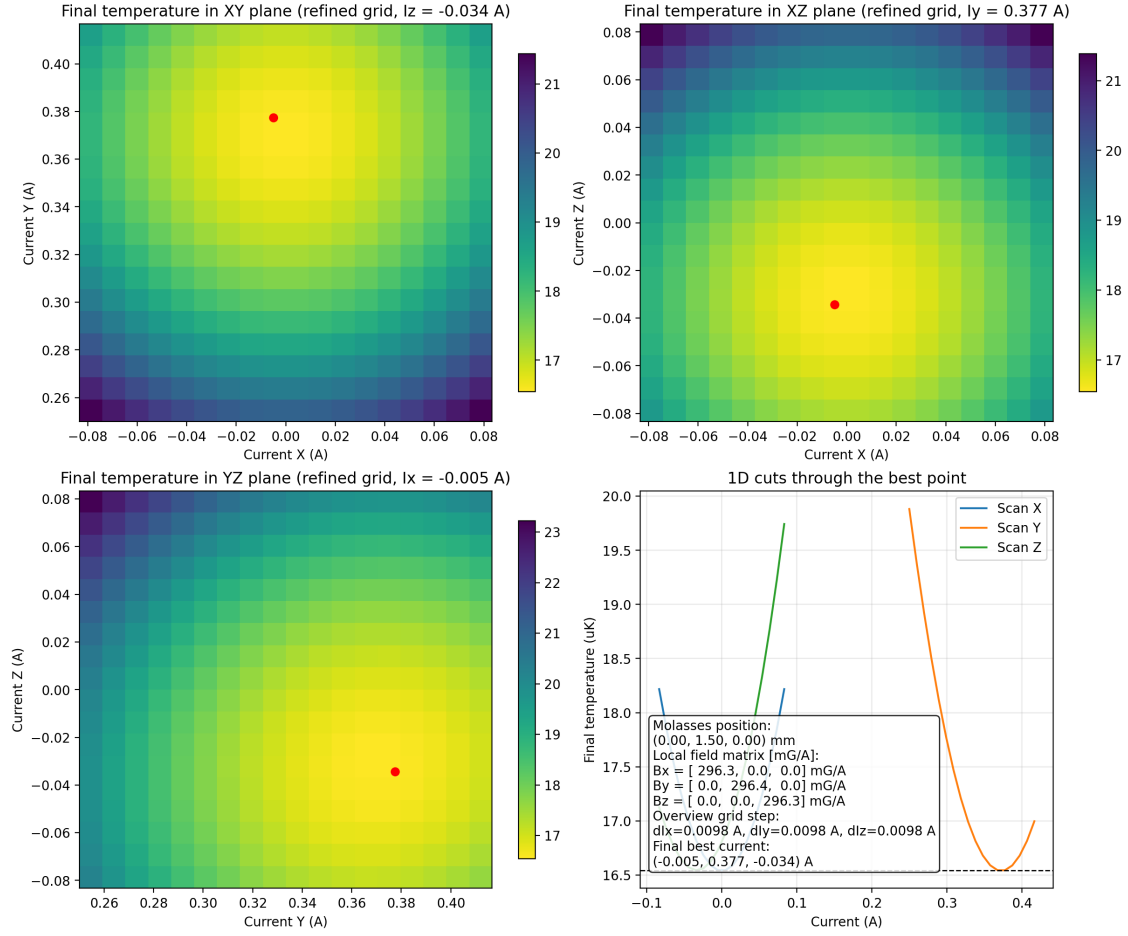


Figure 1: Overview figure generated by the simulation. The three heat maps show final temperature on the XY , XZ , and YZ current slices around the best point. The lower-right panel shows one-dimensional cuts through the optimum.

Figure 1 is the main optimization view. It answers the practical question: “Which compensation current triplet minimizes the final molasses temperature?” The heat maps reveal the temperature landscape in current space, while the one-dimensional cuts clarify the curvature and width of the optimum.

11.2 Time-Domain Dynamics Figure

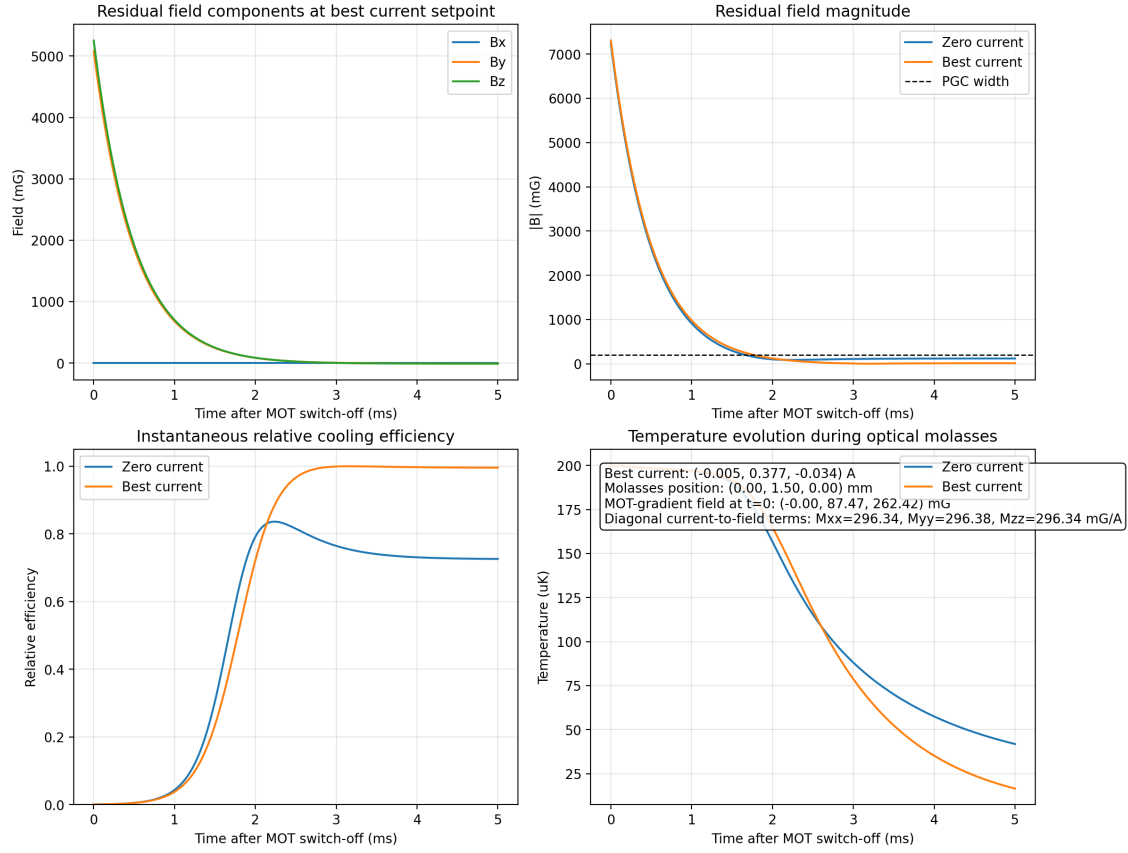


Figure 2: Time-domain dynamics figure. The panels show the vector components of the residual field, the field magnitude, the relative cooling efficiency, and the temperature evolution during molasses.

Figure 2 explains *why* a given current triplet performs better than another one. It shows how the residual field decays after MOT switch-off, how that field maps into the cooling efficiency, and how the temperature approaches its field-dependent equilibrium value.

11.3 Local Spatial Field Figure

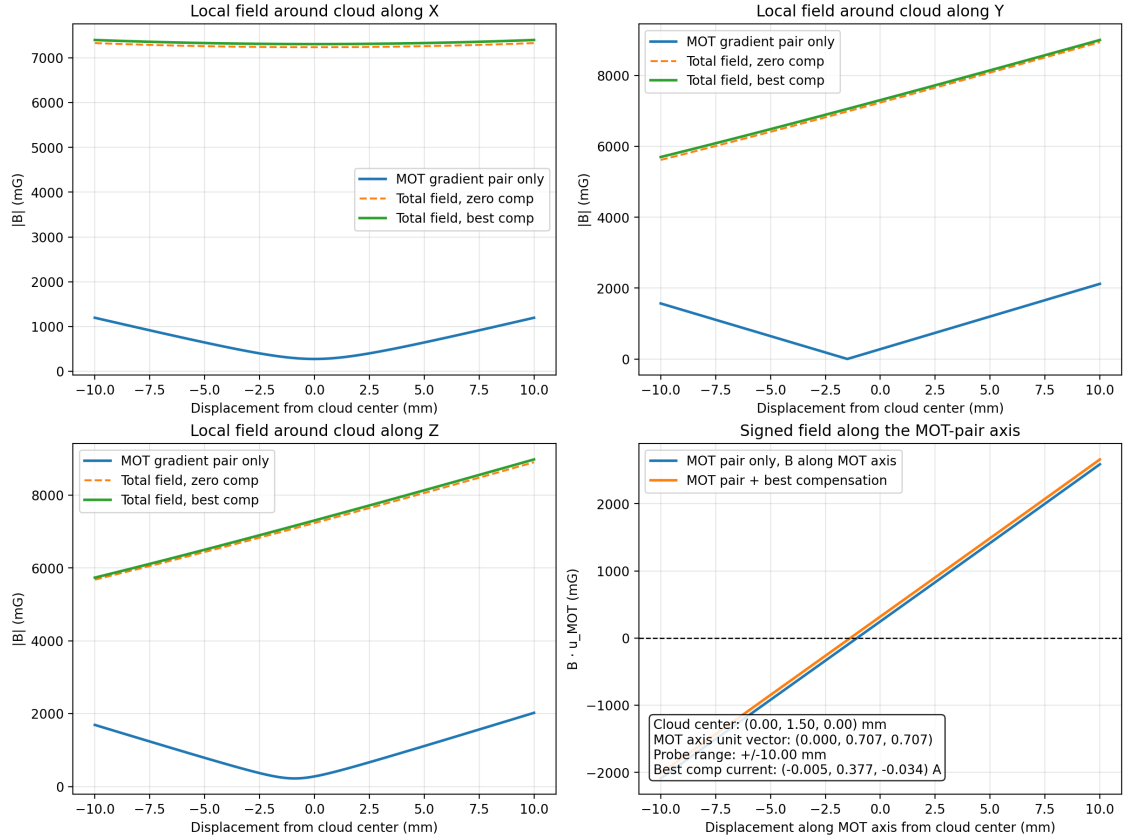


Figure 3: Local spatial magnetic-field figure. The first three panels show field-magnitude cuts around the cloud along X , Y , and Z . The fourth panel shows the signed field along the MOT-pair axis. The figure compares the MOT gradient pair alone, the uncompensated total field, and the total field at the best compensation current.

Figure 3 is especially important once the cloud is displaced from the origin. It visualizes how the local quadrupole field from the 3D MOT coils changes across the cloud neighborhood and how much that profile is improved by the compensation coils. This figure is the clearest diagnostic of whether the compensation is only pointwise or whether it also improves the local field homogeneity.

12 Configuration and Output Files

Each simulation run typically produces:

- `*_scan.csv`: coarse scan table,
- `*_refined_scan.csv`: refined scan table,
- `*_overview.png`: temperature maps in current space,
- `*_dynamics.png`: time-domain residual-field and temperature curves,

- `*_spatial_field.png`: local spatial magnetic-field curves around the cloud,
- `*_summary.txt`: summary of the best point and derived quantities,
- `*_resolved_config.json`: full input plus derived matrices and gradients.

The resolved configuration file stores, among other quantities:

- the compensation current-to-field matrix at the chosen cloud position,
- the MOT residual field at the cloud position at $t = 0$,
- the local MOT gradient Jacobian near the cloud center.

13 Graphical User Interface

The Qt and Tk interfaces provide:

- editable molasses parameters,
- editable compensation-coil geometry and current-scan ranges,
- editable 3D MOT gradient-coil geometry and axis direction,
- editable local spatial-probe range,
- interactive overview, dynamics, and spatial-field plots,
- JSON configuration load/save,
- export and browsing of generated figures and data files.

14 Model Scope and Limitations

The current simulation has several strengths:

- it has moved beyond a purely ideal center-point compensation model,
- it explicitly includes the residual effect of the 3D MOT quadrupole field when the cloud is displaced from the zero,
- it provides a local spatial-field diagnostic rather than only a single-point time trace.

However, several limitations should be kept in mind:

- the temperature evolution still uses the field at the cloud center rather than a full spatial average over a finite cloud distribution,
- the switch-off behavior is modeled with a single exponential time constant,
- the PGC / Sisyphus cooling model is semi-empirical rather than derived from a full multilevel optical Bloch treatment.

15 Possible Future Extensions

To make the simulation closer to the experiment, the following extensions would be natural:

- include a finite cloud size and perform three-dimensional spatial averaging,
- introduce a separate decay constant for the MOT gradient residual,
- include installation errors such as tilt, offset, and asymmetry of the compensation and MOT coils,
- fit the model parameters against measured temperature-versus-current data.

16 Conclusion

The physical logic of the code can be summarized in four steps:

1. compute the local magnetic field produced by the compensation coils and the 3D MOT gradient coils,
2. build the time-dependent total residual field during optical molasses,
3. map that field into cooling efficiency, equilibrium temperature, and cooling time,
4. perform a current scan with optional refinement to locate the best compensation point.

In this sense, the code is useful both as an optimization tool for bias-coil currents and as a diagnostic tool for understanding why a displaced atomic cloud may suffer reduced cooling performance due to residual MOT-gradient fields.